CampusHub Service 层类图（优化版）

认证权限 auth

AuthService
«service»
+ register(req): UserDTO
+ login(req): TokenDTO
+ logout(userId): void
+ validateToken(token): boolean

JwtTokenProvider
«component»
+ generateToken(user): String
+ parseUserId(token): Long
+ validate(token): boolean

生成/解析 JWT

注册/登录用户

封禁/解封

后台管理 admin

AdminService
«service»
+ dashboard(): DashboardDTO
+ banUser(userId, reason): void
+ unbanUser(userId): void
+ listUsers(criteria): Page<UserDTO>

认证审核

处理工单

用户中心 user

CertificationService
«service»
+ submit(req): CertificationDTO
+ getMyCertification(userId): CertificationDTO
+ review(id, decision): void

更新认证状态

UserService
«service»
+ getProfile(userId): UserDTO
+ updateProfile(req): UserDTO
+ getPublicInfo(userId): UserPublicDTO

更新信用分

CreditService
«service»
+ checkCredit(userId): boolean
+ addCredit(userId, amount, reason)
+ deductCredit(userId, amount, reason)
+ listRecords(userId): Page<CreditRecord>

评价影响信用

信用门槛/扣分

订单事件通知

即时通讯 chat

ConversationService
«service»
+ createConversation(req): ConversationDTO
+ listConversations(userId): List<ConversationDTO>
+ sendMessage(req): MessageDTO
+ listMessages(conversationId): Page<MessageDTO>
+ markConversationRead(id): void

新消息提醒

消息通知 notification

NotificationService
«service»
+ createNotification(req): void
+ listNotifications(userId): Page<NotificationDTO>
+ countUnread(userId): long
+ markRead(id): void
+ markAllRead(userId): void

核心业务 experiment

ReviewService
«service»
+ submitReview(req): ReviewDTO
+ listUserReviews(userId): Page<ReviewDTO>
+ getOrderReviews(orderId): List<ReviewDTO>

低分触发仲裁

ArbitrationService
«service»
+ createTicket(req): TicketDTO
+ processTicket(id, decision): void
+ listPending(): Page<TicketDTO>

仲裁进度通知

OrderService
«service»
+ acceptDemand(demandId): OrderDTO
+ markDelivered(orderId): void
+ confirmCompleted(orderId): void
+ cancelOrder(orderId, reason): void
+ autoConfirmExpired(): void

读取/更新需求状态

DemandService
«service»
+ createDemand(req): DemandDTO
+ updateDemand(id, req): DemandDTO
+ cancelDemand(id): void
+ search(criteria): Page<DemandDTO>

智能推荐 recommend

匹配标签/用户

RecommendationService
«service»
+ recommendDemands(userId): List<DemandDTO>
+ recommendAcceptors(demandId): List<UserDTO>

分层边界

Controller 只依赖 Service 接口
Service 负责业务编排与跨模块调用
Repository / Entity 位于下层，图中省略实现细节